Gerry Singh

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PROJECTS

TrayMate - AI-driven meal ordering app for care facilities that personalizes dietary compliance

- Engineered secure multi-user workflows and **4 distinct** role-based access levels (**RBAC**) through a production-grade backend using **Spring Boot** and **JWT**.
- Architected and managed a high-integrity 3rd-Normal Form (**3NF**) relational database in **AWS MySQL**, ensuring data consistency across **10+ interconnected tables**.
- Automated deployment and validation through **CI/CD** pipelines, reducing manual testing overhead by **40%** via **GitHub Actions** and automated test suites such as **Cypress** to ensure 100% reliability in critical meal-selection flows.

Glowlight - Raspberry Pi Climate Monitoring System

- Built a real-time climate monitoring system that provides instant visual feedback through the integration of a **DHT11 sensor** and an RGB LED using **Python** and **Raspberry Pi**.
- Programmed a dual-mode UI using **GPIO** state logic to manage **10+ hardware states**, allowing users to toggle between Edit and Sense modes seamlessly.
- Implemented dynamic reporting by mapping potentiometer inputs to digital LCD limits, maintaining **98% accuracy** in hardware-software signal interfacing.

Graphix - Integral Visualization Tool

- Built a **responsive web interface** using **React** and **Bootstrap**, optimizing UX for various device viewports.
- Optimized math-rendering performance through efficient polynomial data processing using React Hooks.
- Designed an intuitive STEM-focused UI featuring interactive sliders and **LaTeX-style** math notation to enhance user engagement for educational use.

EDUCATION

Seattle Pacific University | *Bachelor of Science in Computer Science* Expected Graduation: June 2026

- GPA: 3.89
- Related Coursework: Data Structures & Algorithms, API Design, Systems Design, and Database Management.

CERTIFICATIONS

Microsoft Certified: Azure AI Engineer Associate

WORK EXPERIENCE

Computer Science Teaching Assistant | Edmonds College Jan 2023 - Sep 2024

- Increased student project success rates by **25%** for a cohort of **60+ students** through guided code reviews of Data Structures and Algorithms in Java and C++.
- Accelerated student debugging proficiency through teaching systematic stack trace analysis using **GDB and IDE tools**.
- Conducted **10+ hours** of weekly 1:1 technical mentorship sessions, simplifying OOP and manual memory management for introductory students.

TECHNICAL SKILLS

Languages: Java, C++, SQL, HTML, CSS, JavaScript, R
Developer Tools: AWS, Postman, Git, GitHub, CI/CD, Azure, Docker, Jira, Cypress, Shiny, Plottly
Libraries/ Framework: Spring Boot, Spring Security, MySQL, ReactJS, Bootstrap